FEDERAL BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, ISLAMABAD

INTERMEDIATE (11TH CLASS) EXAMINATION, 2025

CHEMISTRY

Past Paper 2025 • Paper Code: 1HA-I 25005 — 30051 (Version B) / 32051 (Version D)

Retyped faithfully from the original board paper — same questions, no changes.

VERSION B

CHEMISTRY HSSC-I — SECTION – A (Objective Type)

Time Allowed: 25 Minutes | **Maximum Marks:** 17 | **Paper Code:** 30051 (Version B)

Note: Section – A is compulsory. All parts of this section are to be answered on this page and handed over to the Centre Superintendent. Deleting/overwriting is not allowed. Do not use lead pencil.

حصہ اول لازمی ہے۔ اس کے جوابات اسی صفحہ پر دے کر ناظم مرکز کے حوالے کریں۔ کاٹ کر دوبارہ لکھنے کی اجازت نہیں ہے۔ لیڈ پنسل کا استعمال ممنوع ہے۔

Note: Fill the relevant bubble against each question according to curriculum:

ہر سوال کے سامنے دیے گئے کریکولم کے مطابق درست دائرہ کو پر کریں۔

Q1. The third ionization potential value of an element is significantly high as compared to its 1st and 2nd ionization potentials. The element belongs to:

(A) Group–1 (I-A)
(B) Group–2 (II-A)
(C) Group–17 (VII-A)
(D) Group–18 (VIII-A)

Q2. Which pair of elements will most likely form an ionic bond?

(A) C & H
(B) B & Cl
(C) Na & O
(D) H & Cl

Q3. Which statement is FALSE about comparative bond energy values of Hydrogen halide molecules?

(A) H–Cl > H–F
(B) H–F > H–Cl
(C) H–Cl > H–Br
(D) H–Br > H–I

Q4. The number of molecules present in 67.2 dm^3 of He gas at STP is:

(A) 6.02×10^{24}
(B) 1.204×10^{24}
(C) 1.806×10^{24}
(D) 2.4×10^{24}

Q5. What will be the enthalpy of solution of NaCl if ΔH (reverse lattice) is +786 kj/mol and ΔH (hyd) is -783 kj/mol?

(A) +3 kj/mol
(B) −3 kj/mol
(C) +1569 kj/mol
(D) −1569 kj/mol

Q6. The following equilibrium will NOT be shifted in the forward direction by: ($N_2 + 3H_2 \rightleftharpoons 2NH_3 + \Delta$)

(A) Increasing temperature
(B) Increasing pressure
(C) Adding H_2 gas
(D) Removing NH_3

Q7. Aqueous solution of which of the following salts will have pH less than 7?

(A) Na_2SO_4
(B) KNO_3
(C) $CuSO_4$
(D) Na_2CO_3

Q8. Position of an element in periodic table with electronic configuration $1s^2$, $2s^2p^6$, $3s^2p^2$ is:

(A) In Group–1 (I-A)
(B) In Group–2 (II-A)
(C) In Group–13 (III-A)
(D) In Group–14 (IV-A)

Q9. Peroxyacetyl nitrate (PAN) is formed by the reaction of oxides of nitrogen with:

(A) Oxides of sulphur
(B) Hydrogen sulphide
(C) Oxides of phosphorus
(D) Unburned hydrocarbons

Q10. Which of the following halides can reduce H_2SO_4 to H_2S?

(A) NaF
(B) NaCl
(C) NaBr
(D) NaI

Q11. Identify the step in the mechanism of free radical chlorination of ethane: $CH_3 – CH_3 + Cl\bullet \longrightarrow CH_3 – CH_2\bullet + HCl$

(A) Initiation
(B) Propagation
(C) Substitution
(D) Termination

Q12. Identify the product formed when Ethyl Bromide is warmed with NaOH in ethanol:

(A) $CH_2=CH_2$ (B) $CH_3–CH_2–ONa$

(C) $CH_3–CHO$ (D) $CH_2=CH–Br$

Q13. Which reagent is used for the reduction of Carboxylic acids to Alcohols?

(A) $KMnO_4$ (B) $K_2Cr_2O_7$

(C) $LiAlH_4$ (D) H_2/Pt

Q14. The compound which gives Silver mirror test as well as Iodoform test is:

(A) Ethanol (B) Ethanal

(C) Propanal (D) Propanone

Q15. Which pair of organic compounds can be distinguished by using 2, 4–DNPH?

(A) Ethanol and Acetone (B) 1–Propanol and 2–Propanol

(C) Ethanal and Ethanol (D) Ethyl chloride and Ethyl bromide

Q16. Identify correct relationship for determination of half-life of a first order reaction:

(A) $t_{1/2} = 0.693/k$ (B) $t_{1/2} = [A°]/2k$

(C) $t_{1/2} = 1/k[A°]$ (D) $t_{1/2} = 0.0231[A°]/k$

Q17. Which of the following organic compound can be identified by Fehling solution test?

(A) $CH_3–CH_2–OH$ (B) $CH_3–COOH$

(C) CH_3CH_2CHO (D) $CH_3–CH_2–Cl$

> Note: SUPPLEMENTARY TABLE — Atomic No / Symbol / Mass No for elements 1–20 (H to Ca)

SECTIONS – B & C (Subjective Type)

Time Allowed: 2:35 Hours | **Maximum Marks:** 68 | **Paper Code:** 1HA-I 25005 (B)

Q2. (Marks (14 x 3 = 42)) Q.2 Answers the following parts briefly.

(i) [03] Write down electronic configuration of: $_{29}Cu^{+2}$, $_{33}As$, $_{35}Br^{-1}$ — OR (1+2) — What is meant by Catenation? What is its importance in organic chemistry?

(ii) [1.5+1.5] Briefly explain the structure of PCl_3 & NH_4^+ ion on the basis of VSEPR theory. — OR (03) — Write down the chemical tests with equations to identify chlorides, bromides and iodides present in aqueous solution.

(iii) [03] Keeping in view the factors affecting bond energy, Justify the order of bond energies in the following: a) $Cl_2 > F_2$ b) $H_2 > HBr$ c) $N_2 > O_2$ — OR (03) — Describe any three properties of water which are due to hydrogen bonding.

(iv) [03] Write down three differences between limiting and non-limiting reactants. — OR (03) — Write down the reaction of Ethyl Bromide with KOH_(aq), NH_3 and $AgNO_3$.

(v) [03] State any three differences between crystalline and non–crystalline solids. — OR (03) — Briefly describe the terms 'shell', 'sub–shell' and 'orbital'.

(vi) [1.5+1.5] Why Enthalpy of solution for NH_4NO_3 is +25.7 kj/mol (endothermic process) whereas that of $CaCl_2$ is -83.0 kj/mol (exothermic process)? Explain briefly. — OR (1+2) — What is acid rain? Describe its formation with the help of chemical reactions involved.

(vii) [03] Write down any three measures that ensure the preservation of water resources. — OR (1.5+1.5) — How can iodoform test be applied to the alcohols? Describe by giving chemical equations.

(viii) [1.5+1.5] Why aqueous solution of Na_2CO_3 is basic whereas that of NH_4Cl is acidic? Justify by giving chemical equations. — OR (03) — How can Ethyl amine be prepared from: a) An Alkyl halide b) A Nitrate c) An Amide

(ix) [03] Determine the principal energy level and number of valence electrons in the different sub-shells and orbitals of an element present in 3rd period and group-VI A of the periodic table. — OR (03) — Write down the mechanism of base catalysed nucleophilic addition reaction of formaldehyde (HCHO) with HCN.

(x) [03] Describe the use of compounds of sulphur in any three chemical industries. — OR (1x3) — State Markonikoff's rule by giving an example and its mechanism.

(xi) [03] Calculate the volume occupied by $3.011x10^{24}$ molecules of NH_3 gas enclosed in a container at STP. — OR (03) — How is Ethanol prepared from an/a: a) Alkene b) Alkyl halide c) Carboxylic acid

(xii) [03] Briefly describe β – elimination reaction by giving a suitable example. — OR (03) — Write down the reactions of 1– Propanol with: a) $SOCl_2$ b) PCl_5 c) Conc. $H_2SO_4/180°$

(xiii) [03] Calculate Kc for the following reaction at equilibrium at 100°C. $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$ $[SO_2]=0.59$ M $[O_2]=0.05$ M $[SO_3]=0.259$ M — OR (03) — Calculate total entropy change when NaCl dissolves in water under standard conditions: NaCl_(s) ⟶ Na^+_(aq) + Cl^-_(aq) S°(NaCl)=72.1 j/mol/K, S°(Na^+)=321 j/mol/K S°(Cl^-)=56.5 j/mol/K

(xiv) [03] Draw Born-Haber cycle for the formation of KCl. (Calculation is NOT required) — OR (03) — For the reaction $N_2O_4 \longrightarrow 2NO_2$, the concentration of N_2O_4 is reduced from 0.1M to 0.05M in 60 sec. If reaction is 1st order, calculate rate constant for this reaction.

> Note: Attempt the following questions.

Q3. (Marks 1+4+2) Q.3 What is mass spectrometry? Explain the construction and working of a mass spectrometer. How can average atomic mass of an element be calculated from data obtained? — OR (3+3+1) — Explain formation of covalent bonds in O_2 molecule on the basis of molecular orbital theory and valance bond theory. Justify that molecular orbital theory is superior to valence bond theory.

Q4. (Marks 2x3) Q.4 Describe by giving balanced chemical equations and necessary conditions that how a Carboxylic acid can be prepared from: a) $CH_3 – CH_2 – Cl$ b) [$CH_3 – CH – CH_2 – OH$ over CH_3] c) $C_6H_5 – CHO$ — OR (2x3) — Discuss the effect of

following changes on the given reactions at equilibrium: i) Increasing pressure | $C_3H_8 + 5O_2 \rightleftharpoons 3CO_2 + 4H_2O$ ii) Removing SO_3 | $2SO_2 + O_2 \rightleftharpoons 2SO_3$ iii) Decreasing temperature | $CO_2 + \Delta \rightleftharpoons CO + \frac{1}{2}O_2$

Q5. (Marks 2x3) Q.5 How can one identify the functional groups in the following compounds by chemical tests? Elaborate. a) CH_3CHO b) $CH_3 - CH_2 - OH$ c) $CH_3 - CH = CH_2$ — OR (2+2+2) — Why is Ethene more reactive than Ethane? Write down the reactions of Ethene with: a) H_2SO_4/H_2O b) Cl_2/H_2O

Q6. (Marks 1+2+4) Q.6 What is solubility product? Derive an expression for the solubility product of CaF_2. How can the concept of solubility product be applied to predict the precipitation of CaF_2 on mixing aqueous solutions of Ca^{+2} and F^{-1} ions? — OR (07) — Explain the occurrence of a chemical reaction with reference to molecular collisions, activation energy, formation of activated complex and enthalpy changes in the reaction.

> Note: SUPPLEMENTARY TABLE — Atomic No / Symbol / Mass No for elements 1–20 (H to Ca)

VERSION D

CHEMISTRY HSSC-I — SECTION – A (Objective Type)

Time Allowed: 25 Minutes	**Maximum Marks:** 17	**Paper Code:** 32051 (Version D)

> Note: Section – A is compulsory. All parts of this section are to be answered on this page and handed over to the Centre Superintendent. Deleting/overwriting is not allowed. Do not use lead pencil.

حصہ اول لازمی ہے۔ اس کے جوابات اسی صفحہ پر دے کر ناظم مرکز کے حوالے کریں۔ کاٹ کر دوبارہ لکھنے کی اجازت نہیں ہے۔ لیڈ پنسل کا استعمال ممنوع ہے۔

> Note: Fill the relevant bubble against each question according to curriculum:

ہر سوال کے سامنے دیے گئے کریکولم کے مطابق درست دائرہ کو پر کریں۔

Q1. The electronic configuration of an ion with mass number 63 and charge +2 is [Ne] $3s^2$, $3p^6$, $3d^9$. The number of electrons, protons and neutrons present in the atom are:

(A) e=27 p=27 n=36

(B) e=29 p=27 n=36

(C) e=29 p=29 n=34

(D) e=34 p=29 n=63

Q2. A molecule with formula H_2CO has three sigma bond pairs and one pi bond pair in valence shell of central atom (carbon). Its structure is:

(A) Triangular pyramid
(B) Trigonal planar
(C) Bent
(D) Tetrahedral

Q3. 0.5 dm^3 of CO_2 gas is enclosed in a container at STP. The mass of gas is:

(A) 0.49g
(B) 0.98g
(C) 22.4g
(D) 48.6g

Q4. Which molecule possesses stronger London dispersion forces?

(A) n–Butane
(B) n–Pentane
(C) Iso butane
(D) Iso pentane

Q5. Which of the following is the correct relationship for Gibb's free energy?

(A) G = H + TS
(B) G = H × TS
(C) G = H – TS
(D) G = H/TS

Q6. Correct rate equation for given reaction is: $H_2 + Cl_2 \rightleftharpoons$ (sunlight) 2HCl

(A) Rate = $k[H_2][Cl_2]$
(B) Rate = $k[H_2]$
(C) Rate = $k[Cl_2]$
(D) Rate = k

Q7. Which of the following reaction at equilibrium will NOT be affected by change in pressure?

(A) $N_2O_4 \rightleftharpoons 2NO_2$
(B) $CO + 2H_2 \rightleftharpoons CH_3OH$
(C) $N_2 + O_2 \rightleftharpoons 2NO$
(D) $N_2 + 3H_2 \rightleftharpoons 2NH_3$

Q8. If solubility of a salt MX is $2x10^{-2}$ mol/dm^3, what will be the value of Ksp if salt dissociates into one cation (M^+) and one anion (X^-)?

(A) $1x10^{-4}$
(B) $2x10^{-4}$
(C) $4x10^{-4}$
(D) $8x10^{-4}$

Q9. Which of the following decreases across the period?

(A) Atomic radius
(B) Ionization potential
(C) Electron affinity
(D) Electronegativity

Q10. An element M reacts with O_2, H_2O and Cl_2 to form the products M_2O, MOH and MCl respectively. The element is most likely:

(A) Li
(B) Al
(C) Mg
(D) Ca

Q11. Which compound of sulfur is used as fertilizer?

(A) $(NH_4)_2SO_4$
(B) $CuSO_4$
(C) Na_2SO_4
(D) $BaSO_4$

Q12. Identify x and y, in the reaction: $CH_2 = CH_2 \xrightarrow{x} C_2H_5OH \xrightarrow{y} CH_2 = CH_2$

(A) x=H_2O, y=HCl
(B) x=H_2SO_4, y=$LiAlH_4$
(C) x=H_2SO_4/H_2O, y=Al_2O_3
(D) x=$NaBH_4$, y=H_2SO_4

Q13. An alkyl halide is reacted with $AgNO_3$ in ethanol, light yellow ppt insoluble in NH_4OH are formed. Identify the alkyl halide:

(A) $CH_3–CH_2–F$
(B) $CH_3–CH_2–Cl$
(C) $CH_3–CH_2–Br$
(D) $CH_3–CH_2–I$

Q14. What is produced when ethene is reacted with cold, dilute solution of $KMnO_4$?

(A) $CH_3–CH_2–OH$
(B) $CH_3–CHO$
(C) $CH_3–COOH$
(D) $HO–CH_2–CH_2–OH$

Q15. Identify the product Q in given reaction: $CH_3 – CH_2 – CN + 2H_2O \xrightarrow{(HCl)} Q + NH_3$

(A) $CH_3 – CH_2 – OH$
(B) $CH_3 – CH_2 – COOH$
(C) $CH_3 – CH_2 – NH_2$
(D) $CH_3–COOH$

Q16. Butanone can be prepared by oxidation of:

(A) 1–Butanol
(B) 2–Butanol
(C) Butanal
(D) Butanoic acid

Q17. The reduction of $C_6H_5–NO_2$ by Sn/HCl results in the formation of:

(A) C_6H_6
(B) $C_6H_5 – NH_2$
(C) $C_6H_5 – OH$
(D) $C_6H_5 – Cl$

> Note: SUPPLEMENTARY TABLE — Atomic No / Symbol / Mass No for elements 1–20 (H to Ca)

SECTIONS – B & C (Subjective Type)

Time Allowed: 2:35 Hours | **Maximum Marks:** 68 | **Paper Code:** 1HA-I 25005 (D)

Q2. (Marks (14 x 3 = 42)) Q.2 Answers the following parts briefly. (Use of graph paper is not necessary. Candidates can make their own grid on answer book if required)

(i) [1x3] Write down the electronic configuration of an element with atomic number 24 and also of its ion with charge+3. Describe the reason for any discrepancy in electronic configuration of element. — OR (03) — Briefly explain the terms with examples: a) Enthalpy of atomization, b) Lattice energy, c) Electron affinity.

(ii) [03] Draw molecular orbital diagram for N_2 and calculate its bond order. — OR (1+2) — What is dynamic equilibrium? How does it exist between two physical states of a substance? Describe by giving an example.

(iii) [1+2] What is molar volume? Calculate the volume occupied by 2000g of Cl_2 gas at STP. — OR (03) — How are the elements arranged in different blocks of periodic table?

(iv) [03] Why does carbon form a large number of organic compounds? Give three reasons. — OR (03) — Complete the following reactions: i $CH_2 = CH_2 + H_2O + KMnO_4 \longrightarrow$ ii $CH_3 - CH_2 - COOCH_3 + NaOH \longrightarrow$ iii $C_6H_5CHO + 2[H] \rightarrow (NaBH_4)$

(v) [1+2] What is Azimuthal quantum number? Describe its significance. — OR (1.5+1.5) — Explain why: i is the falling drop of water spherical in shape? ii has water high surface tension than Ether?

(vi) [03] Justify the difference in the boiling points of $CH_4(-162°C)$, $HCl(-85°C)$ and $H_2O(100°C)$. — OR (1+2) — What is the Photo chemical smog? Briefly describe its formation.

(vii) [1x3] What is entropy? Describe the increase or decrease in entropy of system with sign in the following reactions: a) $N_2O_4(g) \longrightarrow 2NO_2(s)$ b) $NH_3(g) + HCl(g) \longrightarrow NH_4Cl(s)$ — OR (03) — Ethanol can be converted into propanoic acid in three stage process. $C_2H_5OH \rightarrow (I)$ A $\rightarrow (II)$ B $\rightarrow (III)$ $C_2H_5 - COOH$ Write down the balanced chemical equation to show the conversion in each step.

(viii) [03] Write down three methods by which un-usable water can be converted into clean water. — OR (03) — Write down any three harmful effects of acid rain.

(ix) [03] Write down three methods for the preparation of Ethyl chloride from an alcohol. — OR (03) — How can primary, secondary and tertiary alcohols be distinguished by applying Lucas test?

(x) [03] Briefly explain the factors which affect the relative stability of primary, secondary and tertiary carbocation. — OR (1.5+1.5) — Write down the reactions of Ethyl Amine ($C_2H_5 - NH_2$) with: a) $CH_3CH_2 - Cl$ b) HNO_2 / HCl

(xi) [03] Differentiate between metallic and non–metallic behavior of elements. — OR (03) — For the reaction $2NO_2 + F_2 \longrightarrow 2NO_2F$ Rate= $[NO_2][F_2]$ propose the mechanism of reaction.

(xii) [03] Describe how (n+l) rule and Hund's rule are applied in writing the electronic and orbital configuration of Mn (Z=25). — OR (03) — How does a catalyst affect the rate of a chemical reaction? Explain by drawing a reaction pathway diagram.

(xiii) [03] Calculate concentration of H^+ ion in 1.0 M solution of Formic acid (HCOOH) which ionizes as: $HCOOH \rightleftharpoons H^+ + HCOO^-$, $K_a = 1.8 \times 10^{-4}$ — OR (1.5+1.5) — How can Iodoform test be applied for identification of aldehydes and ketones? Describe by giving chemical equations.

(xiv) [1.5+1.5] Draw position isomers and functional group isomers of $C_3H_7 - OH$ — OR (03) — Why is the ionization of HF decreased when aqueous solution of its salt is added to it? Briefly explain by applying Le-Chatlier's principle.

> Note: Attempt the following questions. (Use of graph paper is not necessary. Candidates can make their own grid on answer book if required)

Q3. (Marks 1+3+3) Q.3 What is hybridization? Write down three differences between sp^3 and sp^2 hybridizations. Explain bonding in BF_3 on the basis of hybridization. — OR (3+4) — Why are alkanes chemically less reactive? Describe the mechanism of free radical chlorination of Ethane to form Ethyl Chloride.

Q4. (Marks 3+3) Q.4 Explain one physical and one chemical method to recognize a system at equilibrium. — OR (2+4) — Write down a chemical reaction between CH_3COOH and CH_3OH with necessary conditions. Also write the mechanism of this reaction, explaining each step.

Q5. (Marks 3+3) Q.5 Explain the relative reactivity of halogens: (i) as oxidizing agents (ii) in their reaction with H_2, by giving suitable examples. — OR (2x3) — Describe the behaviour of 1–propanol and 2–propanol with respect to their: (i) reaction with I_2/NaOH (ii) reaction with HCl/$ZnCl_2$ (iii) reaction with $K_2Cr_2O_7/H_2SO_4$

Q6. (Marks 07) Q.6 300 g Zn is reacted with 300 g HCl. $Zn + 2HCl \longrightarrow ZnCl_2 + H_2$ Calculate the mass of H_2 produced. In an experiment, if 6g H_2 is obtained, calculate percentage yield. (Atomic masses Zn=65.38 g/mol H=1 g/mol Cl=35.5 g/mol) — OR (2+5) — State Hess's law with an example. Apply Hess's law to calculate enthalpy of formation of CH_3COOH. The standard enthalpies of combustion of carbon, hydrogen and acetic acid are −393.5 kj/mol, −285.8 kj/mol and −875 kj/mol respectively.

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